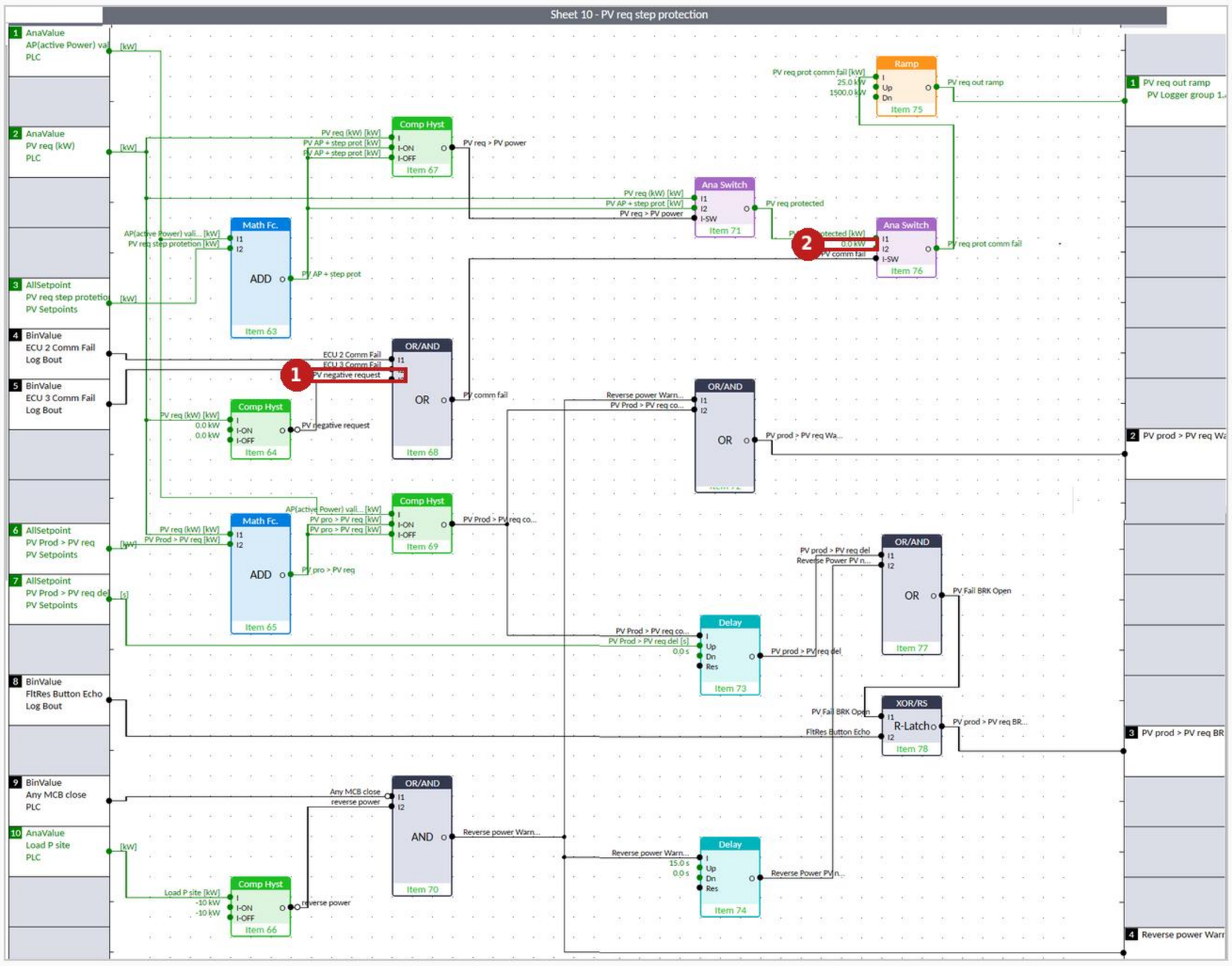
Sheet 10 - PV logger communication: today, why it hurts, and the change

Today, when the controller misses a single reply from the PV logger, the program sends the solar plant a setpoint of zero. The inverters follow it within a few seconds. The gensets suddenly carry the whole plant, the UPS breakers get shed, and then the solar needs about five minutes to climb back, 100 kW at a time. In the history this happens between five and thirty times a day.

The zero does not even reach the logger while the link is down. It only arrives when the link comes back. So it protects nothing, it just costs production every time. The change below makes the program keep its last setpoint instead, and nothing else.

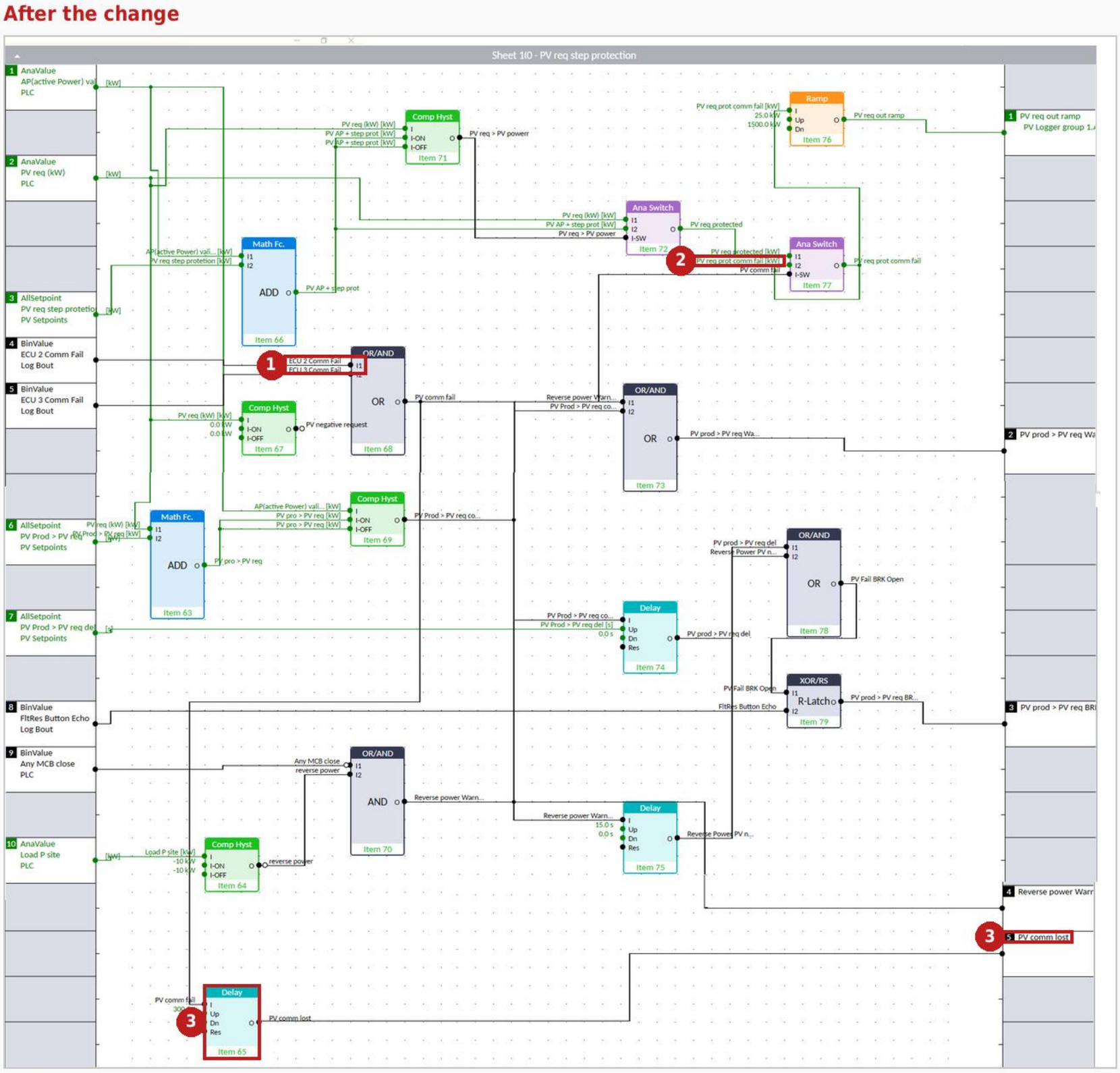
Today



What the program does today

1. Item 68 is the block that decides when the solar request gets cut. It fires when either of the two PV logger connections misses a reply, and also when the request is zero or below.
2. Item 76 is a switch. Its second position is wired to a fixed 0.0 kW. The moment Item 68 fires, that zero is what goes out to the logger. One missed reply, and the whole solar plant is told to stop.

After the change



What it does after the change

1. Item 68 now reacts only to the two real signals, ECU 2 and ECU 3 comm fail. The third input is gone, so a request that is really zero still goes out as zero.
2. The same switch (now Item 77) has its own output on its second input. When a reply is missed it simply keeps sending the value it sent last. The inverters never see a drop, the gensets never see a step, nobody notices.
3. Item 65 is new. It watches the same comm fail signal and, if the link stays dead for five full minutes, raises a flag called PV comm lost (sheet output 5). For now it only feeds an alarm line. Later, if the plant agrees, sheet 3 can use it to open the PV breakers on purpose.

Everything else on the sheet is untouched: the 100 kW step limit, the ramp, the overproduction warning, the reverse power check. With a healthy connection the controller behaves exactly as it does today. If anyone wants it back the way it was: load the archive from 14 October, or put the constant 0.0 kW back on the switch.